

Mechanized Excavation Techniques

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Tunnel Excavation in Rock

It is instructive to consider the fundamental objective of the excavation process - which is to remove rock material (either to create an opening or to obtain material for its inherent value). In order to remove part of a rock mass, it is necessary to induce additional fracturing and fragmentation of the rock.

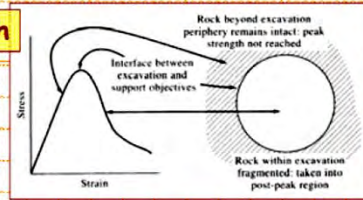
This introduces three critical aspects of excavation:

- ➡ The peak strength of the rock must be exceeded.
- ➡ The *in situ* block size distribution must be changed to the required fragment size distribution.
- ➡ By what means should the required energy be introduced into the rock?



Tunnel Excavation in Rock

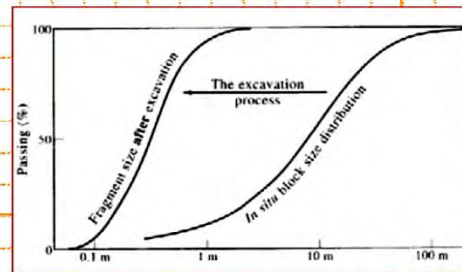
Strength



The tensile strength of rock is about $1/10^{\text{th}}$ the compressive strength and the energy beneath the stress-strain curve is roughly its square. Therefore, breaking the rock in tension requires only $1/100^{\text{th}}$ of the energy as that in compression.

Block Size

The fracturing of rock during excavation changes the natural block size distribution. The goal therefore is to consider how best to move from one curve to the other in the excavation process.



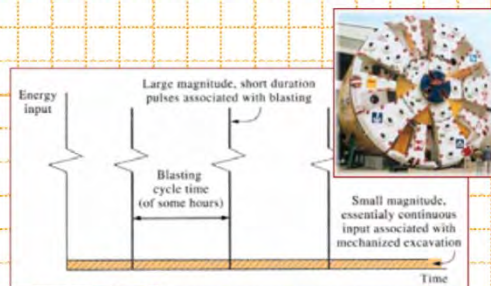
Hudson & Harrison (1997)

Energy and Excavation Process

One objective in the excavation process may be to optimize the use of energy, i.e. the amount of energy required to remove a unit volume of rock (specific energy = J/m^3). There are two fundamental ways of inputting energy into the rock for excavation:

➔ **Blasting:** Energy is input in large quantities over very short durations (cyclical - drill then blast, drill then blast, etc.)

➔ **Machine Excavation:** Energy is input in smaller quantities continuously



Machine Excavation in Rock

There are two basic types of machine for underground rock excavation:



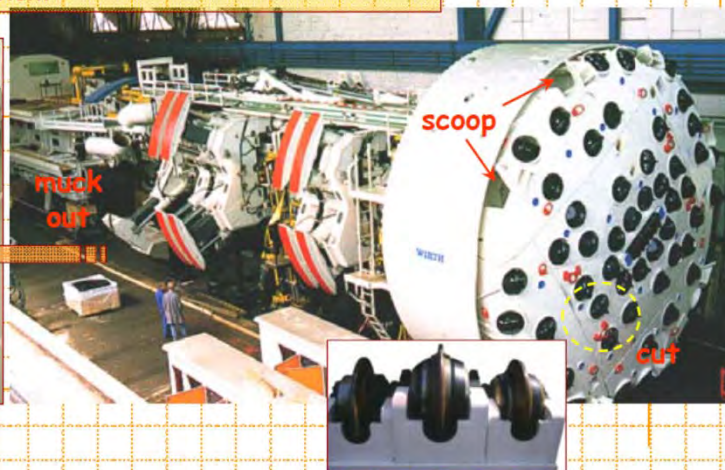
Partial-face machines: use a cutting head on the end of a movable boom (that itself may be track mounted).



Full-face machines: use a rotating head armed with cutters, which fills the tunnel cross-section completely, and thus almost always excavates circular tunnels.

Machine Excavation in Rock

Full-face machines - when used for relatively straight and long tunnels (>2 km) - permit high rates of advance in a smooth, automated construction operation.

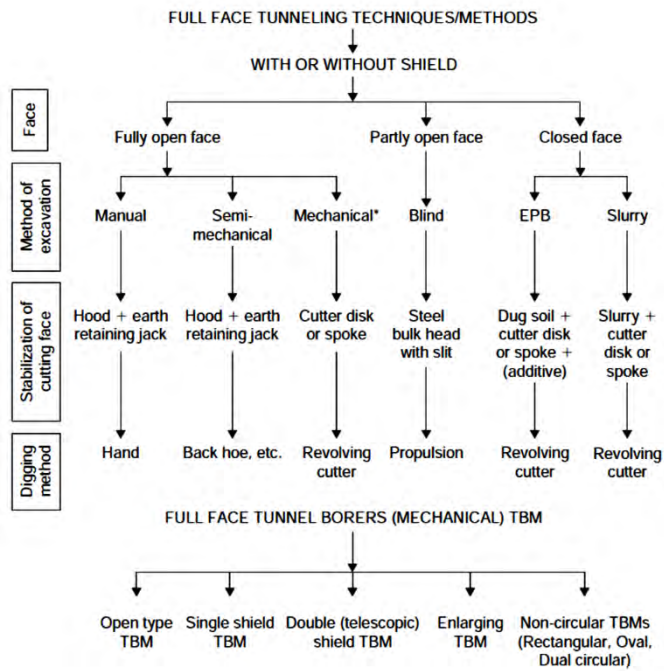


Classification of Underground mechanical excavator by rock type

Hard rock	Soft rock
1.Roadheader 2.TBM 3.Mobile Miner 4.Raise Borer	1.Roadheader 2.Continuous Miner 3.Shearer

Comparison of mechanical excavation with drill and blast systems

Advantages	Disadvantages
• Generally gives more rapid advance than blasting in soft to medium-hard rock. • Fewer unit operations • Smoother excavation profiles • No blasting damage to surrounding rock • Inherently safer	• Cannot excavate very hard rock • High capital cost • Require specialist maintenance • Not as flexible



FULL FACE EXCAVATION

HARD ROCK TUNNELLING

Contrary to soft-ground tunneling where the main objective is to control and support the ground, the goal of hard-rock tunneling is to cut the rock as fast as possible. Daily advance rates of 170 m (diameter 3,4m) have been reported. The application range is extensive and compressive strengths up to 300 MPa can be handled. The diameter range of available TBMs extend from 1.6m - 12m.

The tunnel length should take into consideration the investment costs including as to whether a new or refurbished TBM should be used. A TBM's life time (including some overhauls) is up to 25 km.

FULL FACE EXCAVATION

HARD ROCK TUNNELLING

Long and small tunnels can be driven effectively by Tunnel Boring Machines, TBMs; short and large tunnels (such as highway tunnels) often are more suited to D&B, where permitted.

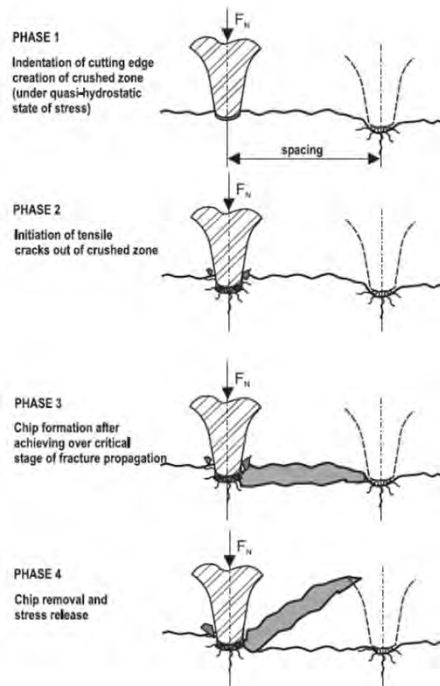
The cutting tool used on TBMs are important. Starting with relatively small discs (< 14" dia) off at approx. 19". There is a tendency today to focus on 17" high performance discs which provide sufficient service life and which keep the design of the TBM in feasible limits.

Basic operation of a TBM

The cutting process is performed by disc cutters. These cutters - generally steel rings - are pressed against the face. The contact pressure between the disc and rock pulverizes the rock on contact and induces lateral cracking towards the neighboring kerf - and rock chipping.

To achieve the best performance, kerf spacing (distance between two adjacent tracking cutters) and cutterload must have suitable values for each rock type.

Average values of 80 - 110 mm spacing and 250 kN cutter load for 17" (= 430 mm) discs are sufficient in most cases.



Cutting process TBM

Based on the experience from many recent TBM-projects, following are the two issues that need mention:

1. Rock anisotropy, such as foliation and/or bedding, can have a significant effect on TBM performance. The Brazilian tensile strength test can provide a reliable indication of the degree and extent of rock anisotropy. This information together with punch penetration and/or fracture toughness tests can be used to develop an assessment of anisotropy influence on TBM performance.
2. The importance of emphasizing the most relevant factors in petrographic (thin section) analysis.

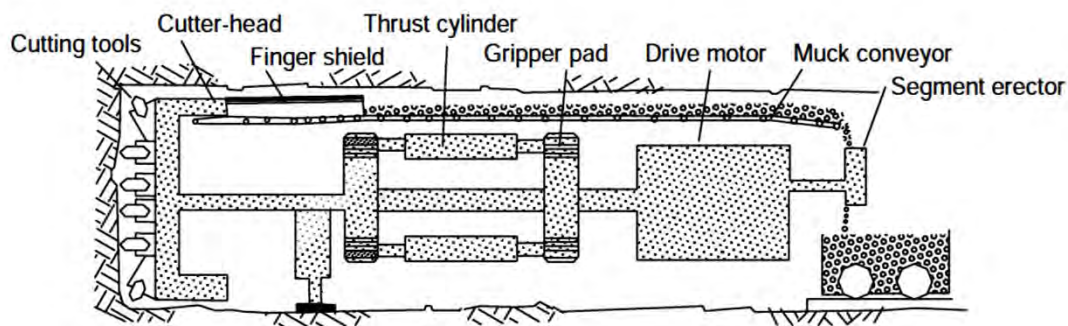
2. The importance of emphasizing the most relevant factors in petrographic (thin section) analysis.

factors of key importance for evaluation of TBM performance and cutter wear, such as:

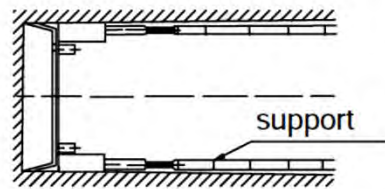
- Grain suturing/interlocking.
- Micro fractures.
- Orientation, directionnel properties.
- Grain size/shape/elongation.
- Content of particular hard minerals (such as quartz, garnet and epidote).
- Any other unusual microscopic features.

Some factors whether not properly investigated or often underestimated still cause the basis for claims in TBM projects, as follows:

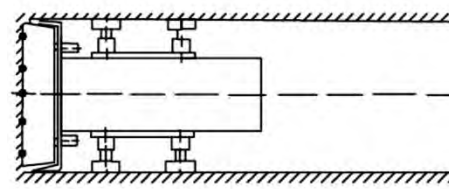
- *Ground water*, which in cases of excessive inflow, can cause great problems and considerable extra cost.
- *Rock stresses*, which in worst case may cause the TBM to get stuck, particularly in squeezing ground.
- *Adverse ground conditions*, such as running or swelling ground.
- *Geologic features*, such as joints, fractures, bedding/foliation, which can have a significant impact on TBM performance.
- *Microscopic features* of the rock such as grain suturing/interlocking, which can increase the difficulty of excavation. Based on experience, particular attention for future projects should be paid on these factors, although the others are also important, and definitely should not be omitted.



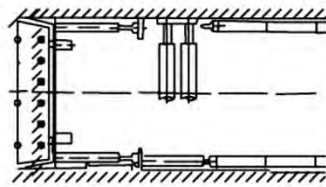
Components of a full face TBM



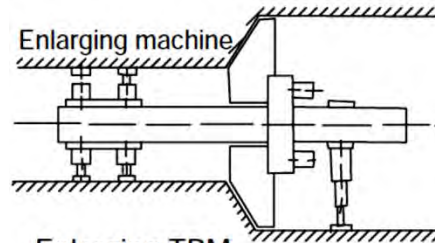
Single shield TBM



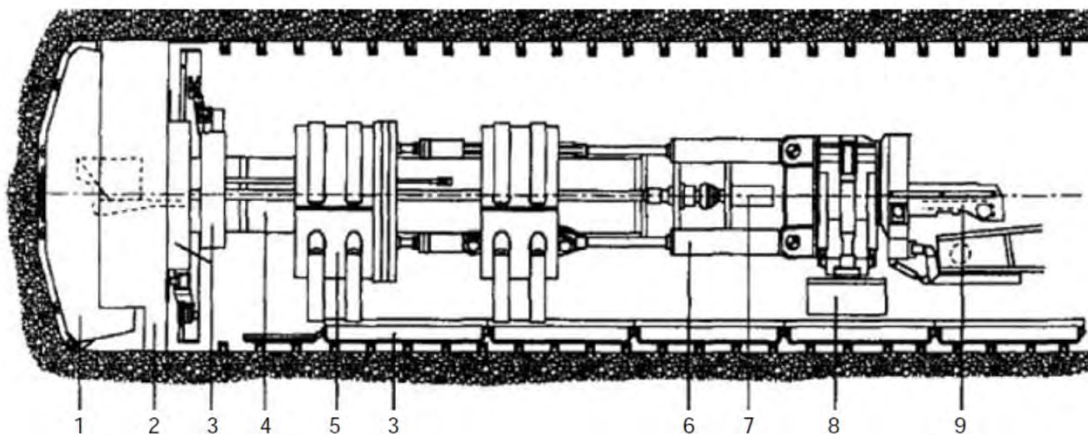
Full face open TBM



Double (telescopic)
shield TBM



Enlarging TBM



1 – Cutter head. 2 – Cutter head shield with support cylinder. 3 – Lining installation and transporting system. 4 – Inner Kelly. 5 – Cutter Kelly with gripper shields and adjustment cylinders. 6 – Advance cylinder. 7 – Cutter head drive. 8 – Rear support. 9 – Belt conveyor.

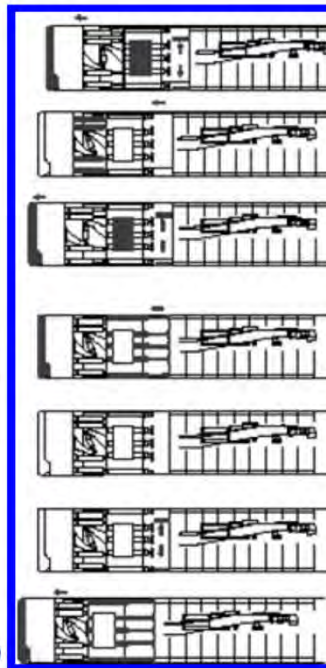
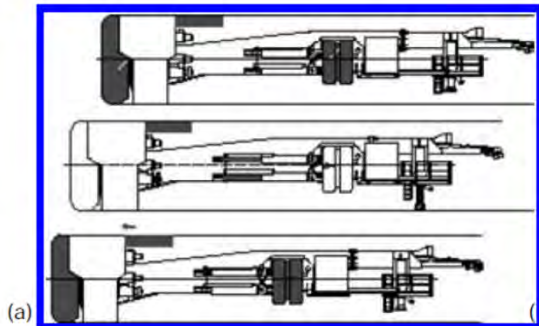
Wirth's TBM and its components

(a) Robin's Single shield TBM working Cycle.

Top – Initial position;
Middle – Re-grip;
Bottom – Bore again.

(b) Robin's Double shield TBMs boring cycle.

(Courtesy: Robbins)



Double shield mode

- 1) Bore & line
- 2) Regrip
- 3) Bore & line again

Boring cycle: shielded TBM

Single shield mode

- 1) Bore
- 2) Retract
- 3) Line
- 4) Bore again

Machine Excavation

The advance rate at which the excavation proceeds is a function of the cutting rate and utilization factor (which is the amount of time that the machine is cutting rock). Factors contributing to low utilization rates are difficulties with ground support and steering, the need to frequently replace cutters, blocked scoops, broken conveyors, etc.



The cutters may damage if the TBM is pushed forwards with too much force, or large blocks fall and strike them.

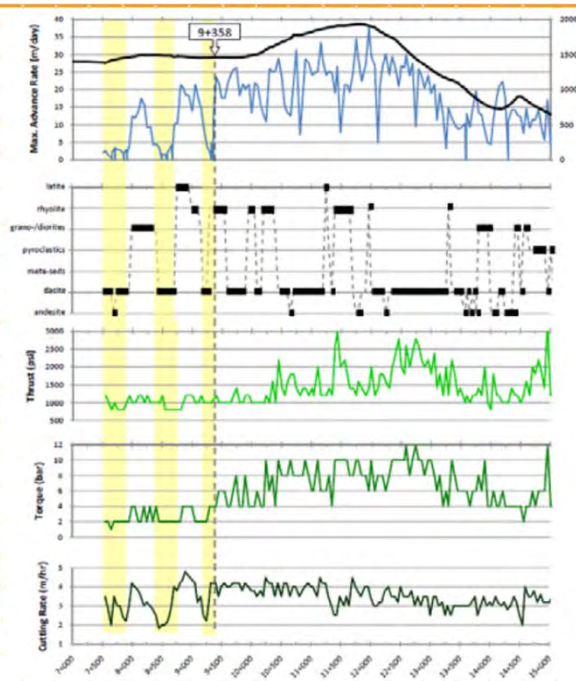
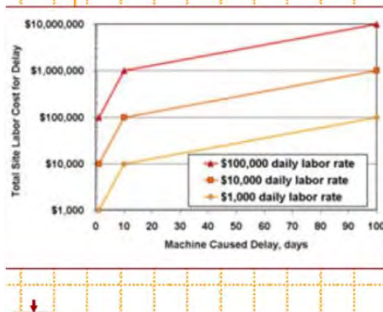


Broken conveyor

TBM Operation

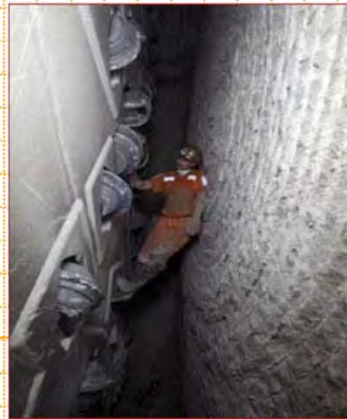
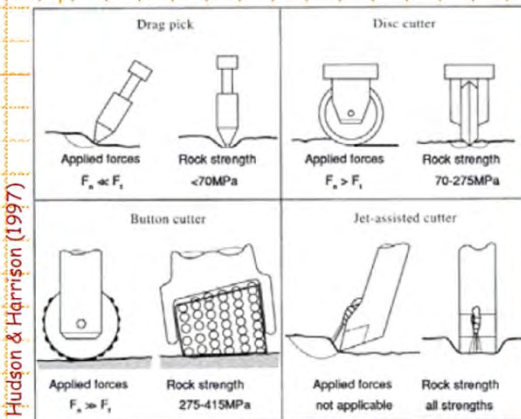
Factors that may control TBM performance include:

- TBM Penetration Rate (meters/machine hour)
- TBM Downtime (minutes)
- TBM Utilization (machine hours/shift hours)
- Tool Wear (tool changes per shift)



Mechanics of Rock Cutting

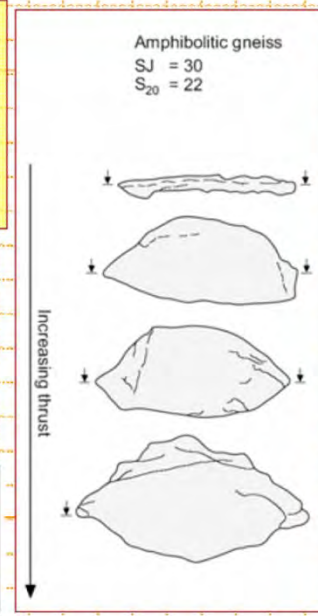
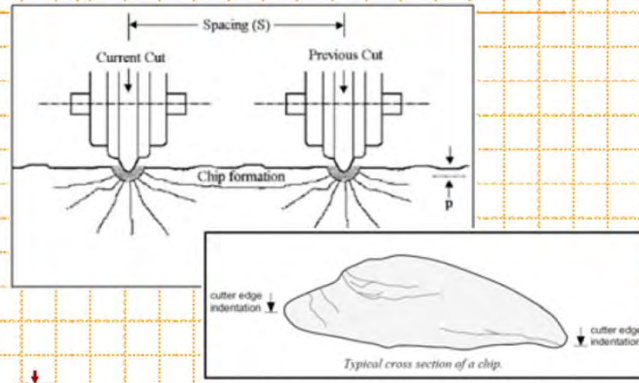
In tunnelling terms, a TBM applies both thrust (F_n) and torque (F_t) during the cutting process. In selecting the proper cutting tool, the engineer wishes to know how the tools should be configured on a machine cutting head, how to minimize the need to replace cutters, how to avoid damaging the cutter mounts, and how to minimize vibration.



Hudson & Harrison (1997)

Mechanics of Rock Cutting

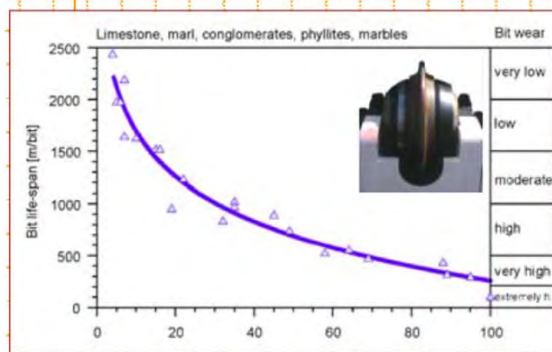
Cutting involves a complex mixture of tensile, shear and compressive modes of failure. With thrust, the cutting disc penetrates the rock and generates extensive crack propagation to the free surface. Further strain relief occurs as the disc edge rolls out of its cut, inducing further tensile cracking and slabbing at the rock surface.



NTNU-Anleggsdrift (1998)

Mechanics of Rock Cutting - Cutter Wear

The primary impact of disc wear on costs can be so severe that cutter costs are often considered as a separate item in bid preparation. In general, 1.5 hours are required for a single cutter change, and if several cutters are changed at one time, each may require 30-40 minutes. Even higher downtimes can be expected with large water inflows, which make cutter change activities more difficult and time-consuming.



Mechanical Excavation - Cutter Heads

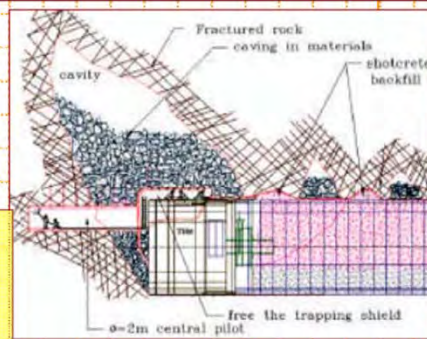
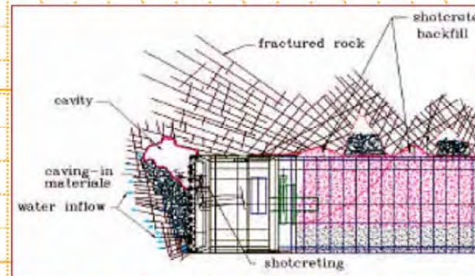
Delays: When the tunnel boring machine is inside the tunnel, the cutters must be changed from the inside the cutting head.



TBM Excavation & Design

The two main factors that will stop tunnel boring machines are either the rock is too hard to cut or that the rock is too soft to sustain the reactionary force necessary to push the machine forward. TBM's will operate within certain ranges of rock deformability and strength, where the machine can be tailored to a specific range to achieve maximum efficiency (the risk being if rock conditions diverge from those the TBM is designed for).

Instability problems at the tunnel face, encountered during excavation of the 12.9km long Pinglin tunnel in Taiwan.



Barla & Pelizza (2000)

TBM Excavation & Design

Table 1 : General classification scheme for tunnelling machines (AITES / ITA, Working Group No.14).

Support			Excavation		Reaction Force	Machine		
Location	System		Method	Tool		Category	Type	
	Cavity	Face						
None			Partial Face Excavating Machines (PFM)	Various	None or Grippers	Rock Machines	Special Rock Tunnelling Machines - Mobile Miner - Continuous Miner - Other	
Cavity	None		Full Face Rotating Cutting Head (TBM)	Cutting disk	Grippers		Unshielded TBM	
				Cutting disk/ Cutting bits/ Cutting knives & teeth	Thrust Jacks		Special Unshielded TBM	
Cutting disk				Grippers and Thrust Jacks	Double Shielded TBM (DS-TBM)			
PFM				Rod header/ Back hoe/ Manual excavation	Thrust Jacks	Open Shield		
Face and cavity	Shield	Mechanical	TBM	Cutting bits/ Cutting knives & teeth	Thrust Jacks	Soft Ground Machines	Mechanical Supported Closed Shield	
			PFM	Road header/Back hoe			Mechanical Supported Open Shield	
			Compressed Air	TBM			Cutting bits/ Cutting knives & teeth	Compressed Air Closed Shield
				PFM			Road header/ Back hoe/ Manual excavation	Compressed Air Open Shield
		Fluid	Slurry	TBM			Cutting disk/ Cutting bits/ Cutting knives & teeth	Close Slurry Shield - Slurry Shield - SS-Hydroshield
				PFM			Road header/Back hoe	Open Slurry Shield - Special Open - Slurry Shields
			Earth Pressure Balance	TBM			Cutting disk/ Cutting bits/ Cutting knives & teeth	Earth Pressure Balance Shield - EPBS Special EPBS
							None or fluid	None or slurry or Earth Press. Balance

TBM Excavation & Design

Single & Double Shield TBM's - Single-shield TBM's are cheaper and are the preferred machine for hard rock tunnelling. Double shielded TBMs are normally used in unstable geology (as they offer more worker protection), or where a high rate of advancement is required.



TBM design

Two basic TBM design principles are available:

- Single-gripper machines
- Double-gripper machines

Both principles have advantages and disadvantages, single gripper machines are used more frequently in standard tunneling projects.

The single-gripper TBM

The basic concept comprises a main frame with a main drive, a floating support at the front end and a gripper at the rear end, required for transferring induced forces into the tunnel wall. A rotating cutterhead is attached to the main gear and rotates at approx. 2.5 m/s peripheral speed. The cutterhead is thrust forward by advance jacks. After a stroke of 1.5 - 1.8, the machine must regrip for a new stroke.

The front support is provided by a dust shield, which is a steel structure with expendable plates in the upper area and a rigid support in the lower area. It seals off the working area and makes dust collection easier. This front support is kept in frictional contact with the tunnel wall and overthrust by the installed thrust force.

The single-gripper TBM

The machine is steered by adjusting the rear end of the frame and turning the machine around the front support. A single-gripper machine can be steered continuously during the boring operation which results a smooth surface in the tunnel. Careful steering only while the head is rotating is essential so as to avoid gage cutter and main bearing damage. The curve radius of the TBM is approximately >150 m, and < 100 m in special designs.

A belt conveyor handles muck discharge. It is installed in the main frame and loaded by buckets on the cutterhead via a hopper in the center of the cutterhead. For maintenance reasons and cutter change, the belt can be retracted to give access to the rear inside area of the cutterhead. The belt discharges into the main conveyor which leads through the back up and discharges into the muck train on the back-up.

The operator's cabin can be placed on the TBM or the back-up, depending on tunnel requirements, which influence the back-up design.

Double-gripper TBMs

Contrary to the single-gripper machine, the double-gripper TBM is supported by two sets of grippers that perform the whole guiding function of the TBM. The front dust shield only seals off the dust from the tunnel and cleans the invert.

The main frame, which is stabilized by the grippers, does not move. To advance the cutterhead, a sliding inner frame is used. Steering during boring is almost impossible; and therefore double-gripper TBM's bore a polygonal tunnel line. Muck discharge is also done by a belt conveyor from the top of the frame to the end of the TBM.

Double grippers have the advantage of better distributing the gripper forces to the tunnel wall in weak ground. However a disadvantage is taking up free work space for passage and consolidation projects at least in smaller diameters. Furthermore the skewed process stresses gage cutters and main cutterhead bearing.

Shielded TBMs

For special conditions in which core sampling is impossible, or the rock is known to be very weak and fractured and if the contract also specifies partial or continuous concrete lining, the shielded TBM is the right equipment for the job. The shielded TBM, as the name says, looks similar to a shield, but the working process is different.

A shielded TBM is a hard-rock TBM enclosed by a shielded body. The rear end of the machine has a pair of integrated grippers to stabilize the TBM in the tunnel. The front end with the cutterhead is pushed forward out of the telescopic shield via advance jacks. A ring of segments can be simultaneously erected under the shield tail cover. After completing the cutting stroke and segment lining erection simultaneously, the TBM's rear part with the grippers is reset into the next position.

Shielded TBMs

If the rock is too weak to give enough resistance for advancing the TBM, the shield mode can be used. In this case, the rear thrust jacks will push against the segment ring and advance the machine. In this event, a parallel operation is not possible which slows the advance rate.

Segment systems used together with shielded TBMs usually serve as a primary lining; and is not watertight. Honeycomb or normal lining can be used. Developments of watertight lining systems have been developed, but provide at the time only sufficient tightness of low pressure conditions. Daily advance rates of up to more than 100 m can be achieved but require excellent logistics from the jobsite organization. The backup system performs similar requirements as for open TBMs in addition with segment handling and grouting logistics.



Example of a shielded TBM.

Full Shield TBM

Suitable for broad range of ground conditions

Because of a cylindrical shell structure, the Full Shield TBMs can be suitable for broad range of ground conditions, from tough rocks to soft strata including fracture zones.

Large internal space

The cylindrical shell structure secures large space of the internal center of TBM, which enables safety maintenance and efficient operation.

Variety of discharge systems

The best discharge system can be selected from variety of discharge systems, such as belt conveyor, screw conveyor, slurry discharge and jet pump transportation according to diverse ground conditions.



φ5m full shield TBM(30 mR, belt conveyor discharge system)



30 mR, articulate 6'

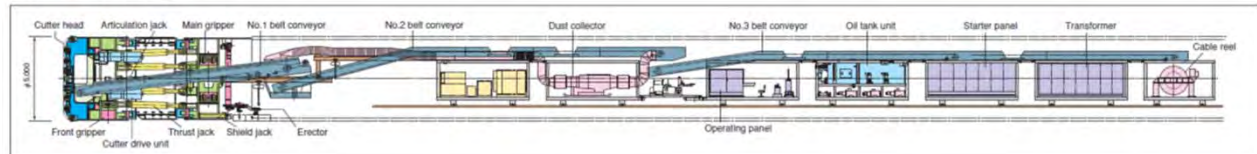


30 mR excavation



Break through

■ Full shield TBM layout



TBM Excavation & Design

Impacts of Geotechnical Conditions on TBM Operations

Major Geotechnical Conditions

Loosening loads, blocky/slabby rock, overbreak, cave-ins

Consequences/Requirements

At the face: cutterhead jams, disc impact loading, cutter disc and mount damage possible, additional loss on available torque for cutting, entry to the face may be required with impact on equipment selection, recessed cutters may be recommended for face ground control.

In the tunnel: short stand-up time, delays for immediate and additional support (perhaps grouting, hand-mining), special equipment (perhaps machine modifications), gripper anchorage and steering difficulty, shut-down in extreme cases of face and crown instability. Extent of zones (perhaps with verification by advance sensing/probe hole drilling) may dictate shield required, and potential impact on lining type selection (as expanded segmental linings may not be reasonable), grouting, and backpacking time and costs may be high.

Groundwater inflow



Low flow/low pressure - operating nuisance, slow-down, adequate pumping capability
high flow and/or high pressure - construction safety concerns, progress slow or shut-down, special procedures for support and water/wet muck handling, may require advance sensing/probe hole drilling.

Corrosive or high-salt water - treatment may be required before disposal, equipment damage, concrete reactivity, problems during facility operation.

Equipment modifications (as water-proofing) may be required if inflow is unanticipated - significant delays.

U.S. Army Corps of Engineers (1997)

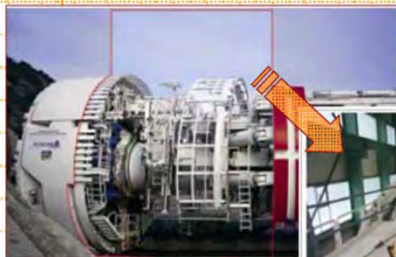
TBM Excavation & Design

Impacts of Geotechnical Conditions on TBM Operations

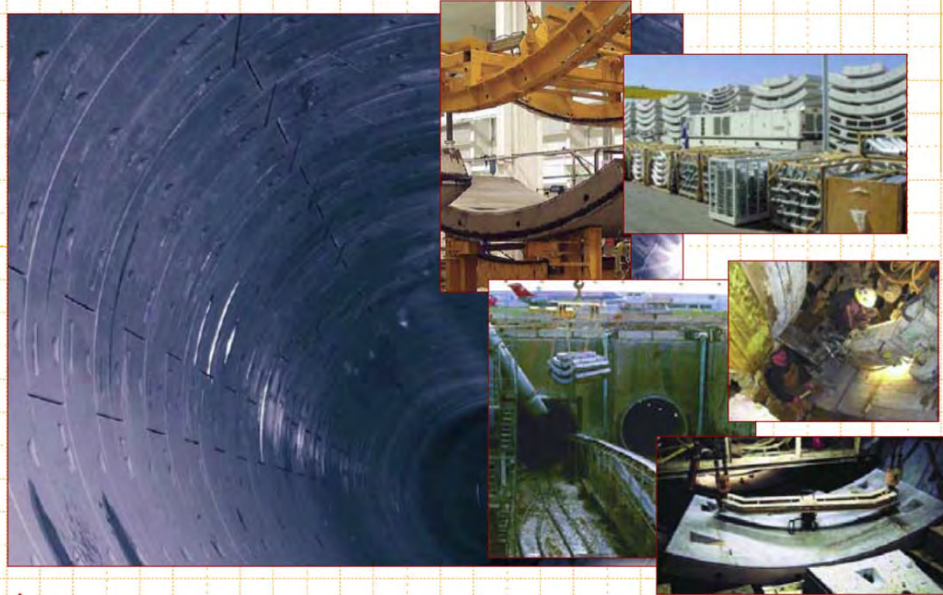
Major Geotechnical Conditions	Consequences/Requirements
Squeezing ground	Shield stalling, must determine how extensive and how fast squeeze can develop, delays for immediate support, equipment modifications may be needed, if invert heave and train mucking - track repair and derail downtime.
Ground gas/hazardous fluids/wastes	Construction safety concerns, safe equipment more expensive, need increased ventilation capacity, delays for advance sensing/probing and perhaps project shut-down, special equipment modifications with great delays if unanticipated, muck management and disposal problems.
Overstress, spalls, bursts	Delays for immediate support, perhaps progress shut-down, construction safety concerns, special procedures may be required.
Hard, abrasive rock	Reduced PR_{rev} and increased F_n - TBM needs adequate installed capacities to achieve reasonable advance rates, delays for high cutter wear and cutterhead damage (especially if jointed/fractured), cutterhead fatigue, and potential bearing problems
Mixed-strength rock	Impact disc loading may increase failure rates, concern for side wall gripping problems with open shields, possible steering problems.
Variable weathering, soil-like zones, faults	Slowed progress, if sidewall grippers not usable may need shield, immediate and additional support, potential for groundwater inflow, muck transport (handling and derailed) problems, steering difficulty, weathering particularly important in argillaceous rock.
Weak rock at invert	Reduced utilization from poor trackability, grade, and alignment - steering problems.

U.S. Army Corps of Engineers (1997)

TBM Excavation & Design



TBM Excavation & Design - Pre-Cast Linings



Tunnelling Breakthroughs



Sequential Excavation & Design - Benches

Bench excavations are used for large diameter tunnels in weak rock. The benefits are that the weak rock will be easier to control for a small opening and reinforcement can be progressively installed along the heading before benching downward. Variations may involve sequences in which the invert, top heading and bench are excavated in different order.

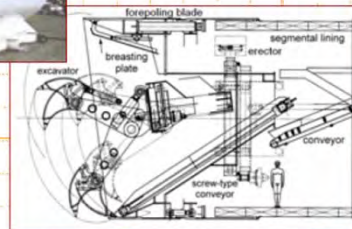


Shield Tunnelling

Advantages of shield tunnelling in soft ground:

1. Tunnel construction can be performed as one step at its full dimensions.
2. Constant support is provided to the advancing tunnel even though it takes the form of a moving system.
3. Omission of temporary support is compensated for by virtue of the immediate installation of the permanent lining.

Whittaker & Frith (1990)



Shield Tunnelling

Open- & Closed-Face Shields - When the tunnel face is free standing and does not require continuous support, the shield is operated in 'Open Mode'. The face is mechanically supported by the cuttinghead while the flood control doors regulate muck flow from the face to the cuttinghead chamber. The excavated muck is rapidly extracted by the conveyor. With a closed-face, an airlock and bulkhead are used to allow the "excavation chamber" to be pressurized with compressed air or a slurry to aid face support.



Open-face shields

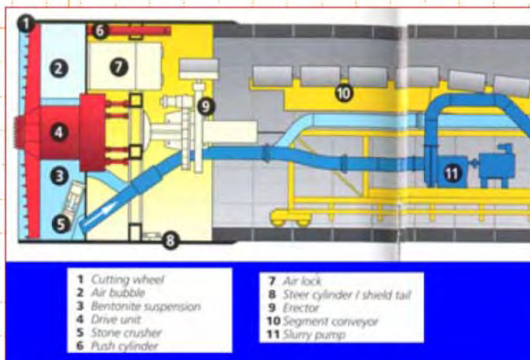
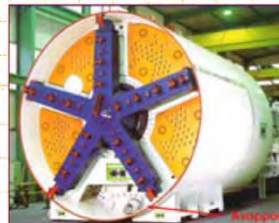


Closed-face shields



Slurry Shields

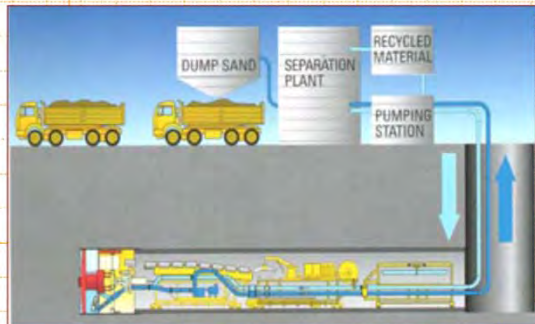
Due to problems regarding health and safety as well as operational aspects (in highly permeable ground, maintaining air pressure at the tunnel face is difficult), compressed air shields are being used less and less. Instead, **slurry shields** and **earth-pressure balance shields** are being favoured more.



Slurry shield operating principle:

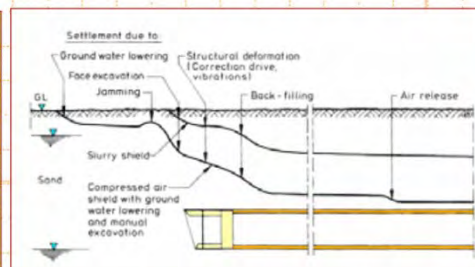
- Tunnel face is supported by bentonite slurry (i.e. tunnel is free from compressed air);
- The slurry is mixed and pumped into a closed excavation chamber;
- The slurry enters the ground, sealing it (filter cake) and enabling pressure to be built up and balanced with the earth and water pressure.

Slurry Shields



As the ground is excavated, it is mixed with the slurry in the excavation chamber. The ground/suspension mixture is then pumped to the surface. In a separation plant, the slurry is separated from the ground. New bentonite is added as required, and the fluid is pumped back to the tunnel face.

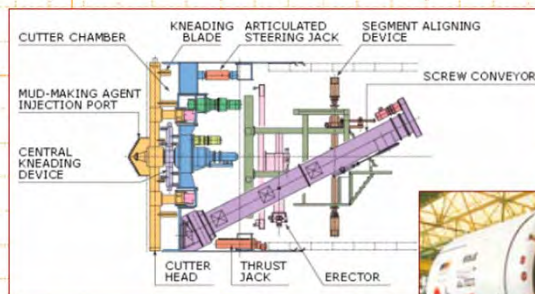
Overall, slurry shields provide a safe tunnelling method causing low settlements. Application is possible in all kinds of loose ground with/without groundwater. Disadvantages include the separation plant (cost, space, energy requirements) and environmental hazards related to tailings (non-separable bentonite slurry containing fines).



Maidl et al. (1996)

Earth Pressure Balance Shields

With a growing percentage of fines, slurry shield tunnelling requires an increasing degree of sophistication and cost for separation (and increasing frequency of slurry renewal). Apart from the high costs and environmental hazards involved, the confined space in most major cities makes the installation of a separation plant on surface difficult. Such were the conditions encountered in the early 70's in Japan, which led to the development of Earth-Pressure Balance shields (EPB).

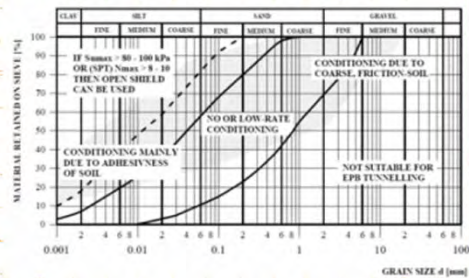


EPB Shields: provide continuous support to the face by balancing earth pressure against machine thrust. As the cutterhead rotates and the shield advances, the excavated earth is mixed with foams in the cutterhead chamber to control its viscosity. The pressure is then adjusted by means of the rate of its extraction (by screw conveyor).

Earth Pressure Balance Shields

The key advantages of EPB Shields are that no separation plant is required and that the method is economically favourable in ground with a high percentage of silt/clay.

clay foam injector



When using EPB tunnelling mode, no bentonite and special treatment plants are necessary and the outcoming soil is nearly natural. If additives like Foam or Polymers are used, highly biodegradable versions exist which can be 95% destroyed after 28 days.

Tunnel Excavation in Soft Ground Conditions

Machine	
Category	Type
Soft Ground Machines	Open Shield
	Mechanical Supported Closed Shield
	Mechanical Supported Open Shield
	Compressed Air Closed Shield
	Compressed Air Open Shield
	Close Slurry Shield - Slurry Shield - SS-Hydroshield
	Open Slurry Shield - Special Open - Slurry Shields
	Earth Pressure Balance Shield - EPBS Special EPBS
	Combined Shield - Mix Shield - Polishield

Open shields are favoured where the ground is free standing.

Closed shields are favoured where the ground is very weak, such as soft clay, silt or running sand.

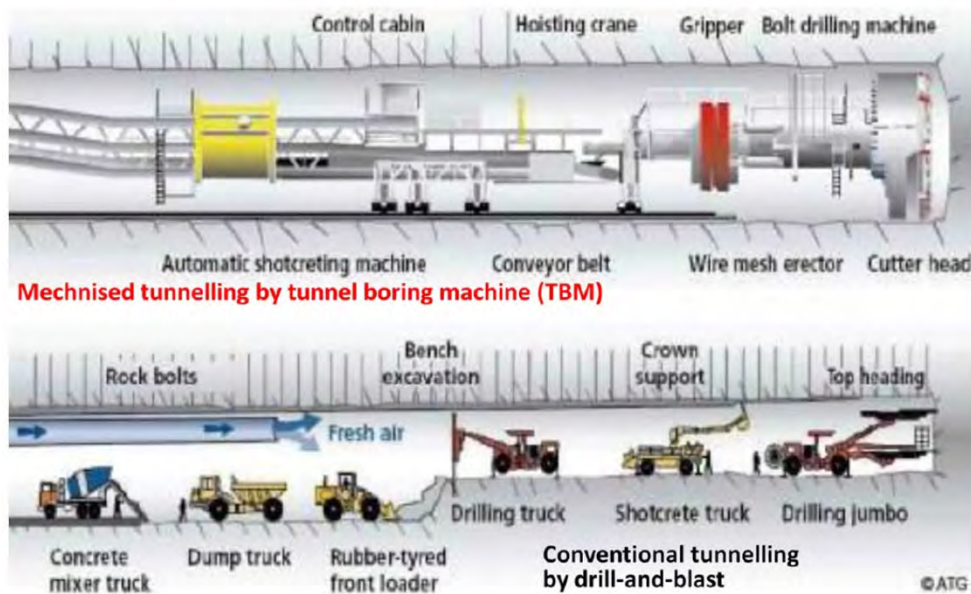
Slurry shields are favoured for water saturated sandy soils and gravels (<10% clay and silt content; e.g. running sand).

EPB shields are favoured for water saturated silty soils (>7% clay and silt content; <70% gravel content).

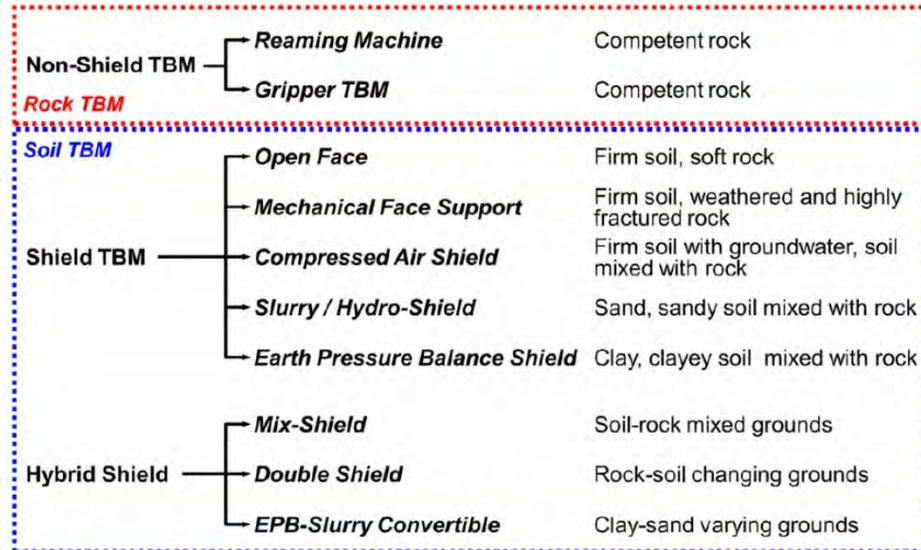
Tunnel Excavation in Soft Ground Conditions

Support			Excavation		Reaction Force	Machine		
Location	System		Method	Tool		Category	Type	
	Cavity	Face						
Face and cavity	Shield			PFM	Rod header/ Back hoe/ Manual excavation	Thrust Jacks	Soft Ground Machines	Open Shield
		Mechanical		TBM	Cutting bits/ Cutting knives & teeth			Mechanical Supported Closed Shield
				PFM	Road header/Back hoe			Mechanical Supported Open Shield
		Fluid	Compressed Air	TBM	Cutting bits/Cutting knives & teeth			Compressed Air Closed Shield
				PFM	Road header/ Back hoe/ Manual excavation			Compressed Air Open Shield
			Slurry	TBM	Cutting disk/ Cutting bits/ Cutting knives & teeth			Close Slurry Shield – Slurry Shield – SS-Hydroshield
				PFM	Road header/Back hoe			Open Slurry Shield – Special Open - Slurry Shields
			Earth Pres- sure Balance	TBM	Cutting disk/ Cutting bits/ Cutting knives & teeth			Earth Pressure Balance Shield - EPBS Special EPBS
								Combined Shield - Mix Shield - Polishield
		None or fluid	None or slurry or Earth Press. Balance					

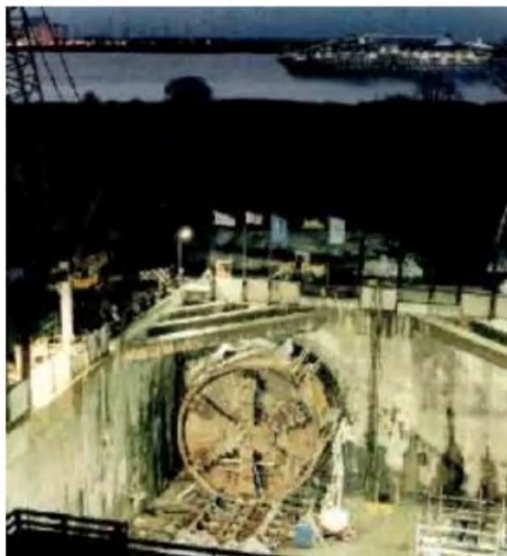
Conventional and TBM Tunnelling



TBM Tunnelling



Basic Functions of TBM

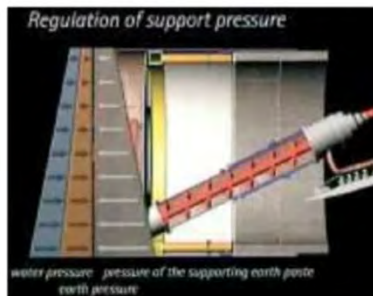


1. Excavating the Ground



Basic Functions of TBM

2. Supporting the Ground



Basic Functions of TBM

3. Mucking the Excavated Materials



Basic Functions of TBM

ADVANTAGES

Enhanced health and safety conditions for the workers.

Industrialization of the tunnelling process, with ensuing reductions in cost and construction times.

Possibility of crossing complex geological and hydrogeological conditions safely and economically.

Good quality of finished product (surrounding ground less altered, precast segment lining).

RISKS

Lack of flexibility: Once the technique has been chosen it is virtually impossible to change it throughout the construction of the tunnel.



Therefore, a correct analysis of different parameters is needed for the choice of the correct mechanized tunnelling technique

Gripper TBM

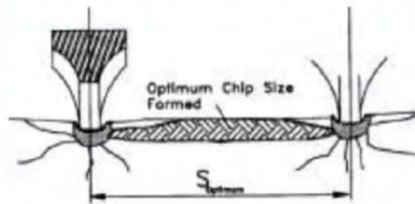
Rock excavation machine for competent and hard rocks.
Movement of machine uses the grippers.



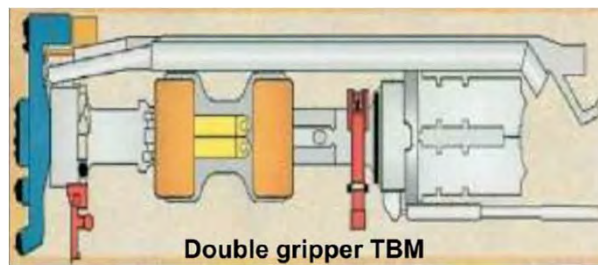
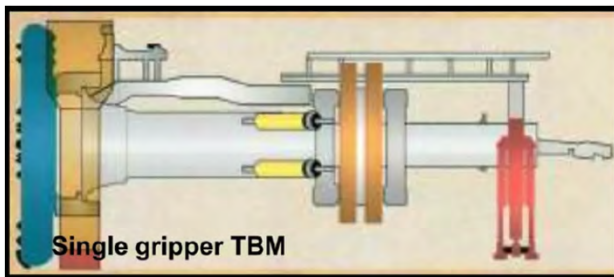
Gripper TBM



Rock is fragmented by fracturing and chipping between cutters.



Gripper TBM



Gripper TBM

ADVANTAGES

Rapid excavation rate in hard rocks.

Capable to excavate hard rocks of UCS up to 300 MPa.

For tunnels of diameter up to 14 m (largest 14.4 m, 2008).

Flexible rock support measure according to ground conditions.

LIMITATIONS

No protection or support given to the rock while excavating. Limited to stable ground.

Gripper may have difficult in poor rock masses with low strength.

Dedusting is needed.

Open Face Shield TBM

Open Face Shield TBM has no face support.

When needed, compressed air pressure can be used to support the face.

It is used limited and only in very consistent firm ground requires no face support.



Slurry / Hydro-Shield

TBM excavation face is supported by pressurizing bentonite in cutterhead chamber. Circulation of the fluid flushes out the muck. Pressure is maintained by controlling discharge rates.



1. Cutting wheel
2. Air bubble
3. Bentonite suspension
4. Drive unit
5. Stone crusher
6. Push cylinder
7. Air lock
8. Steer cylinder/ shield tail
9. Segment erector
10. Segment conveyor

Slurry / Hydro-Shield

ADVANTAGES

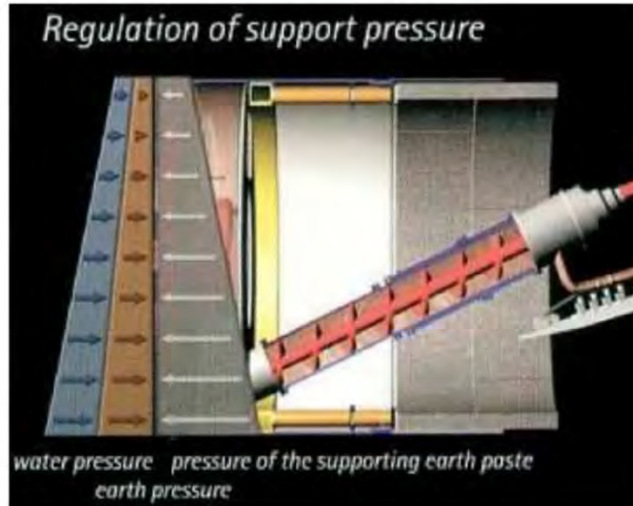
- Equal distribution of pressures against mixed-face conditions.
- Good performance in sands and gravels.
- Gives better face stability (formation of mud-cake).
- Allows access to the face with compressed air.

LIMITATIONS

- Need of large separation plants for bentonite separation and treatment.
- Not suitable for clayey-silty grounds.

Earth Pressure Balance Shield

TBM excavation face is supported by pressurizing soil (earth) inside the cutterhead chamber. Earth pressure in the chamber is regulated by the rate of earth discharge through the screw conveyor.



Earth Pressure Balance Shield



Foams and chemical additives are used to conditioning soil for better control of pressure and muck discharge, in sandy grounds.

Earth Pressure Balance Shield

ADVANTAGES

Very well adapted for excavating silty and clayey grounds.

No need of separation plant.

The use of additives (foams) enables EPB's to be used with sandy-gravelly soils.

LIMITATIONS

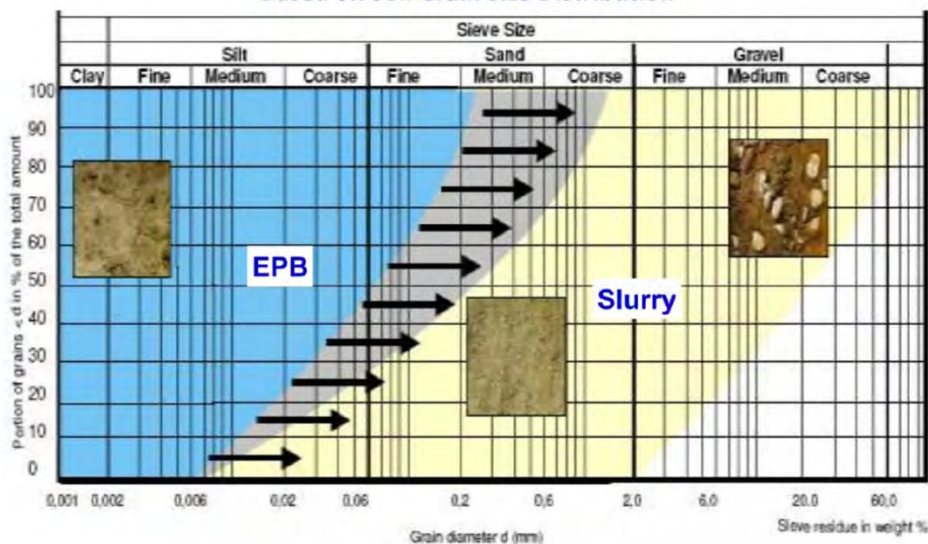
Irregular distribution of pressure against mixed face conditions.

Problems when boulders are encountered: no possibility of fitting a rock crusher.

Wearing of the screw can lead to bad pressure distribution.

Earth Pressure Balance Shield

Selection of Shield Boring Machine Based on Soil Grain Size Distribution



Double Shield

It combines the features of gripper and shield in one TBM, and enables fast excavation even in varying rock formations.



Double Shield

In poor ground condition, it works as a single shield machine. When in rocks, grippers are used for forward movement, so segment can be simultaneously installed while excavating.

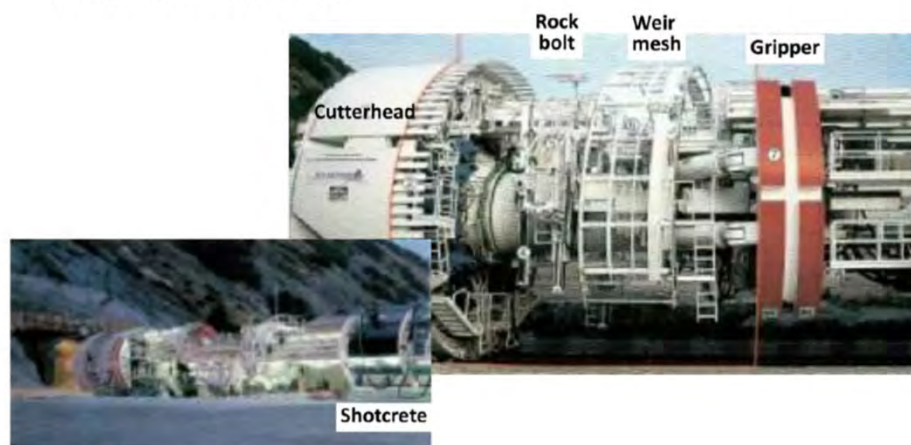


Double Shield

ADVANTAGES	LIMITATIONS
Good flexibility for varying grounds. High advancement rates. The increase depends on operation. Possibility to have different support measures, e.g., segmental lining, and bolts-shotcrete. Possibility of ground treatment and/or probing through the gap between shields.	Maintenance problems of articulation between the shields. Long shield length: possible drawback for squeezing grounds. More expensive than conventional shields.

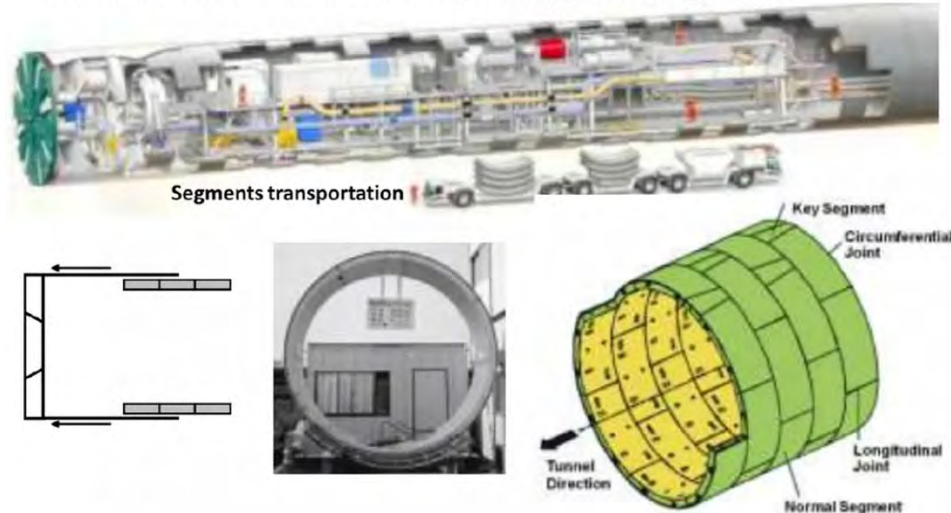
Support of TBM Tunnels

For non-shield TBMs, typical rock support are applied, using bolts, shotcrete, wire mesh or steel sets, as determined by the ground condition.



Support of TBM Tunnels

For shield TBMs, precast concrete segments are erected inside the shield to form a ring to support the ground.



New TBM Technologies

e-Tunnelling and Smart TBM

- Focus on the use of information technology in tunnelling;
- Instrumentation and real-time monitoring of ground condition, ground movement, TBM reaction;
- Data and information management to guide the best operation of TBM;
- Future toward self-learning and smart TBM.